

Pitching Assessment

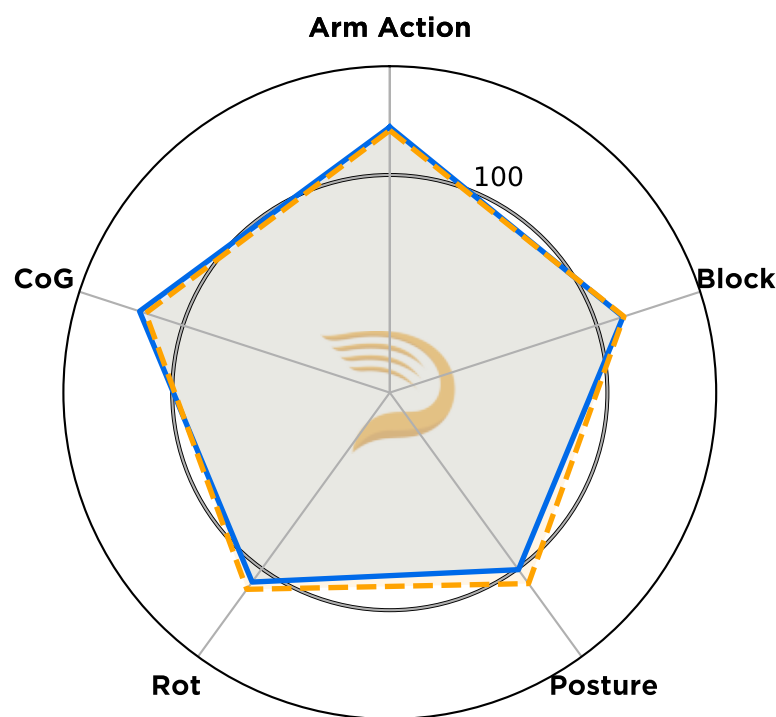


Mechanical Composite Scores⁺ 2024-12-30

93.2 mph (+12.6_{AE})

Est. Mechanical Ceiling: 94.6 mph

Arm Action:	122 (+2)
Block:	113 (-0)
Posture:	101 (-8)
Rot:	108 (-4)
COG:	121 (+3)
Total:	116 (-3)



— 2024-12-30 (93.2mph)
- - 2024-09-05 (89.9mph)

Mechanical Composite Scores quantify mechanical skill separate from physical development. Average scores grade **100. (Higher = better)**

The total score is a weighted composite of the individual scores; weighted by each area's correlation to performance.

Est. Mechanical Ceiling extrapolates performance above expected to **150** grade mechanics.

Est. = Estimated; AE = AboveExpected; Rot = Rotation; CoG = CenterofGravity

CoG (120 / +5.1 mph)



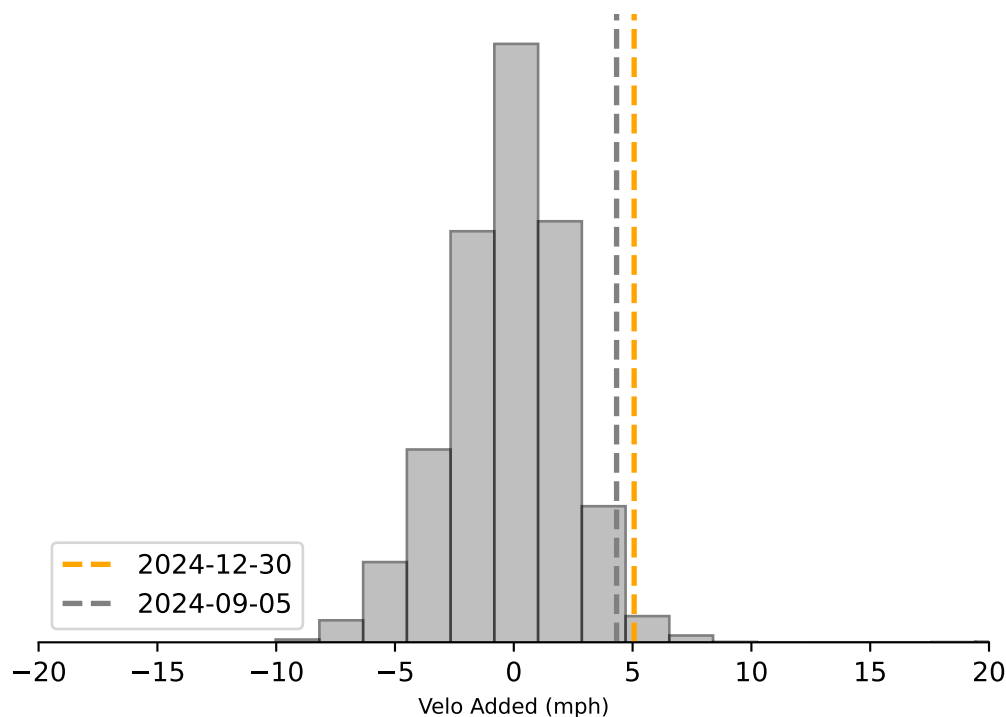
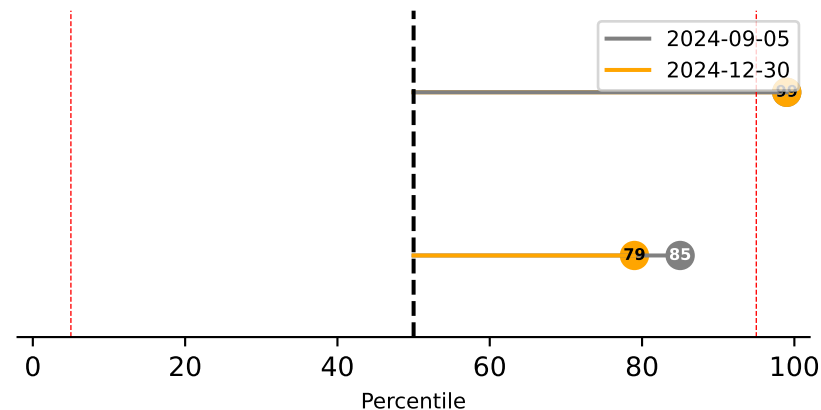
The CoG model uses metrics related to the athlete's center of mass to provide a one-number estimate of an athlete's delivery pace; how quickly and efficiently the athlete moves their entire body down the mound.

CoG is the fourth most impactful model for velocity & fifth most impactful for velocity above expected.

The model summary is outlined below.

Per1mph is the estimated feature change needed, **on average**, for a 1mph change in velocity above expected.

Feature	Athlete	Median Elite	Importance	Per 1mph
CoG Decel	2.23 m/s (+0.11)	1.61 m/s	high	0.15 m/s
Max CoG Velo	3.02 m/s (-0.04)	2.84 m/s	med	0.35 m/s



Top Left: model summary & athlete metrics. If a baseline session is on file, metric change from previous assessment is shown in parenthesis. **Median Elite** is the 50th percentile value for each metric across all elite pitchers (90+ mph).

Top Right: athlete metric percentiles. Dashed vertical lines represent the 5th, 50th, & 95th percentiles.

Left of the thick dashed line = lower than average elite pitcher. Right of thick dashed line = higher than average elite pitcher.

Bottom Left: Total velocity added across all model features. Provides an estimate of how much velocity is added by the athlete's cog, compared to the average cog score

Posture (100 / -0.0 mph)

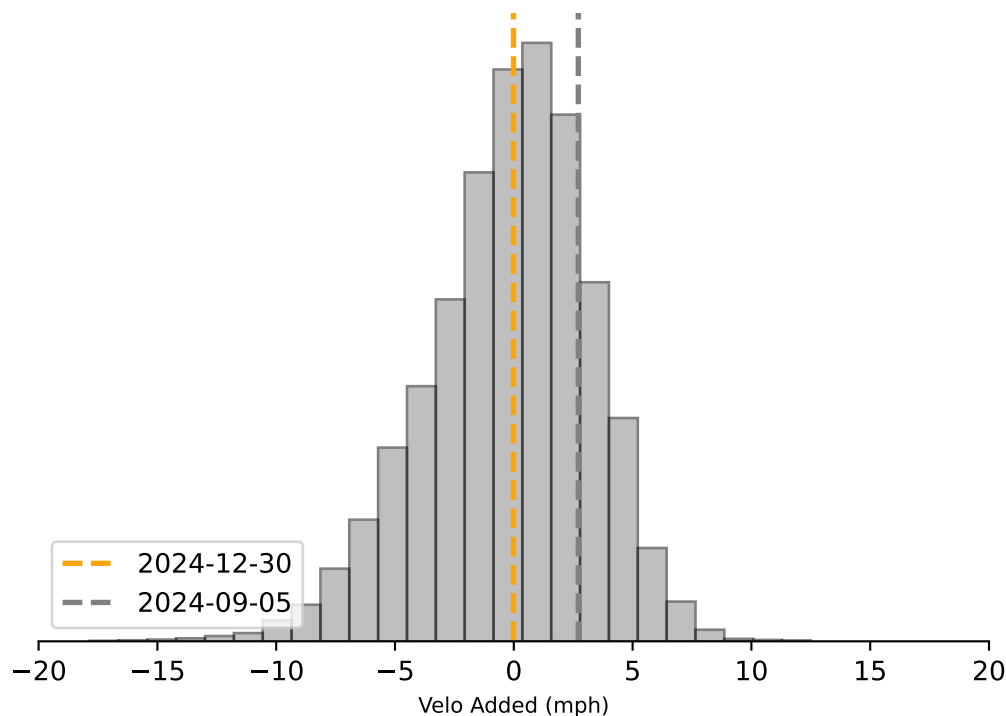
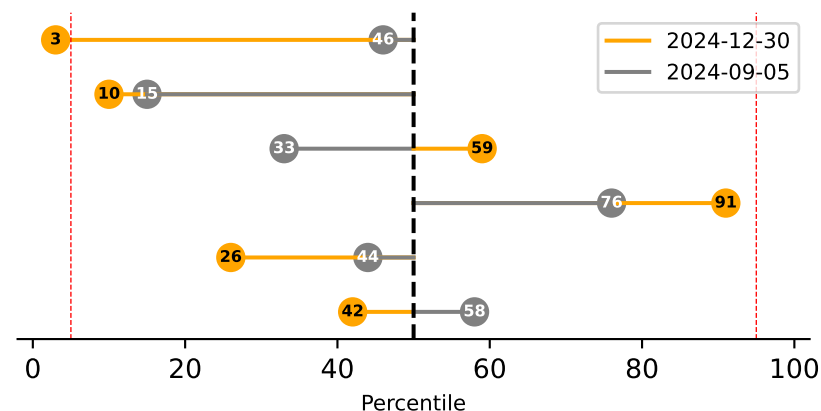


The posture model provides a one-number estimate of how well the athlete maintains their posture through foot plant and into release. Posture is the most impactful model for velocity & second most impactful for velocity above expected.

The model summary is outlined below.

Per1mph is the estimated feature change needed, **on average**, for a 1mph change in velocity above expected.

Feature	Athlete	Median Elite	Importance	Per 1mph
Peak Hip-Shoulder Sep at FP.	15 deg (-15)	30 deg	high	3 deg
Peak Torso Counter Rot.	-51 deg (-3)	-38 deg	high	13 deg
Torso Forward Tilt at FP	5 deg (+5)	4 deg	high	6 deg
Torso Rotation at FP	15 deg (+7)	2 deg	high	10 deg
Torso Side Bend at MER	19 deg (-4)	25 deg	med	3 deg
Torso Rotation at BR	110 deg (-3)	111 deg	med	6 deg



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Bottom Left: Total velocity added across all model features. Provides an estimate of how much velocity is added by the athlete's posture, compared to the average posture score

Block (112 / +3.2 mph)

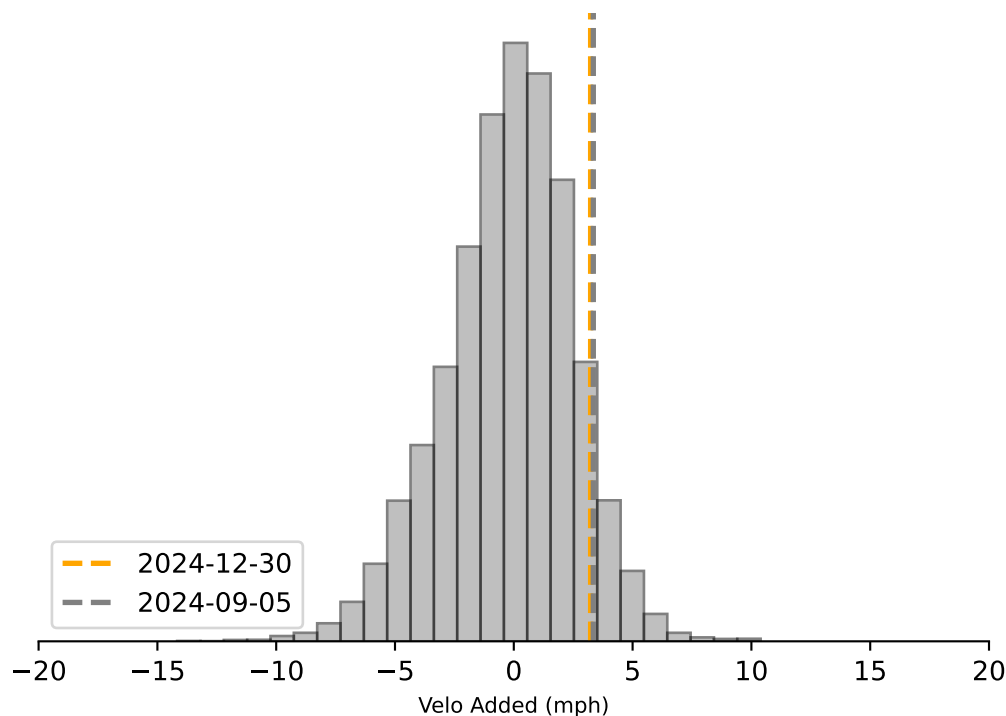
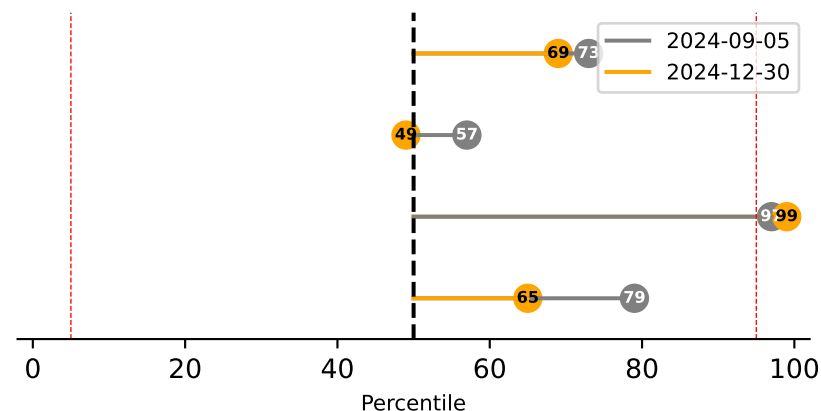
The block model uses lower body metrics to provide a one-number estimate of how well the athlete uses their lead leg to block forward momentum during the stride-to-throwing phase transition.

Block is the fifth most impactful model for velocity & fourth most impactful for velocity above expected.

The model summary is outlined below.

Per1mph is the estimated feature change needed, **on average**, for a 1mph change in velocity above expected.

Feature	Athlete	Median Elite	Importance	Per 1mph
Lead Knee Extension	17 deg (-2)	11 deg	high	5 deg
Stride Length	58 in (-1)	58 in	high	5 in
CoG Decel. AE	0.52 m/s (+0.13)	0.02 m/s	med	0.35 m/s
Peak Lead Knee Ext. Velo	391 deg/s (-85)	316 deg/s	low	103 deg/s



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Bottom Left: Total velocity added across all model features.

Provides an estimate of how much velocity is added by the athlete's block, compared to the average block score

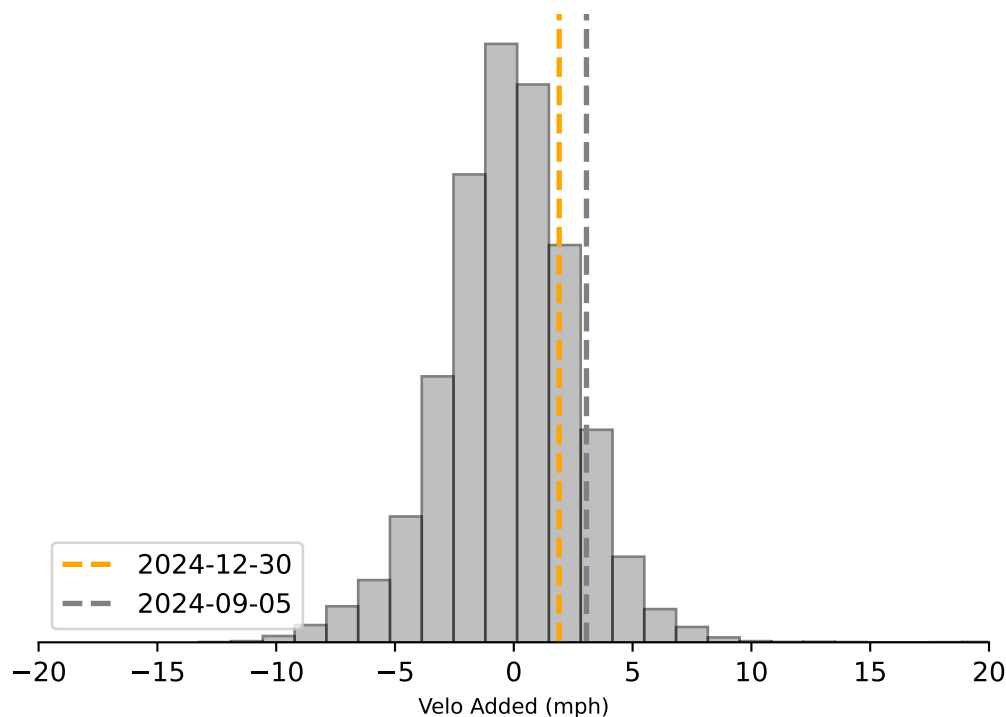
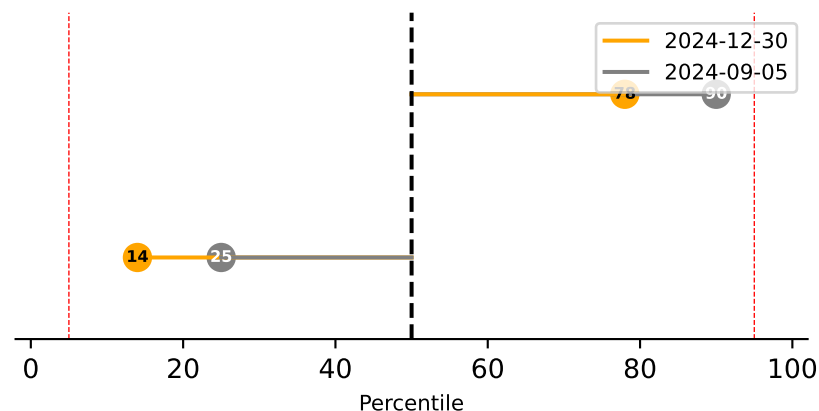
Rotation (107 / +1.9 mph)

The rotation model provides a one-number estimate of how well the athlete rotates their trunk. Rotation is the the fourth most impactful model for velocity & third most impactful for velocity above expected.

The model summary is outlined below.

Per1mph is the estimated feature change needed, **on average**, for a 1mph change in velocity above expected.

Feature	Athlete	Median Elite	Importance	Per 1mph
Torso Rotation Velo	1030 deg/s (-45)	965 deg/s	high	40 deg/s
Pelvis Rotation Velo	515 deg/s (-27)	597 deg/s	low	128 deg/s



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Bottom Left: Total velocity added across all model features.

Provides an estimate of how much velocity is added by the athlete's rotation, compared to the average rotation score

Arm Action (122 / +7.5 mph)

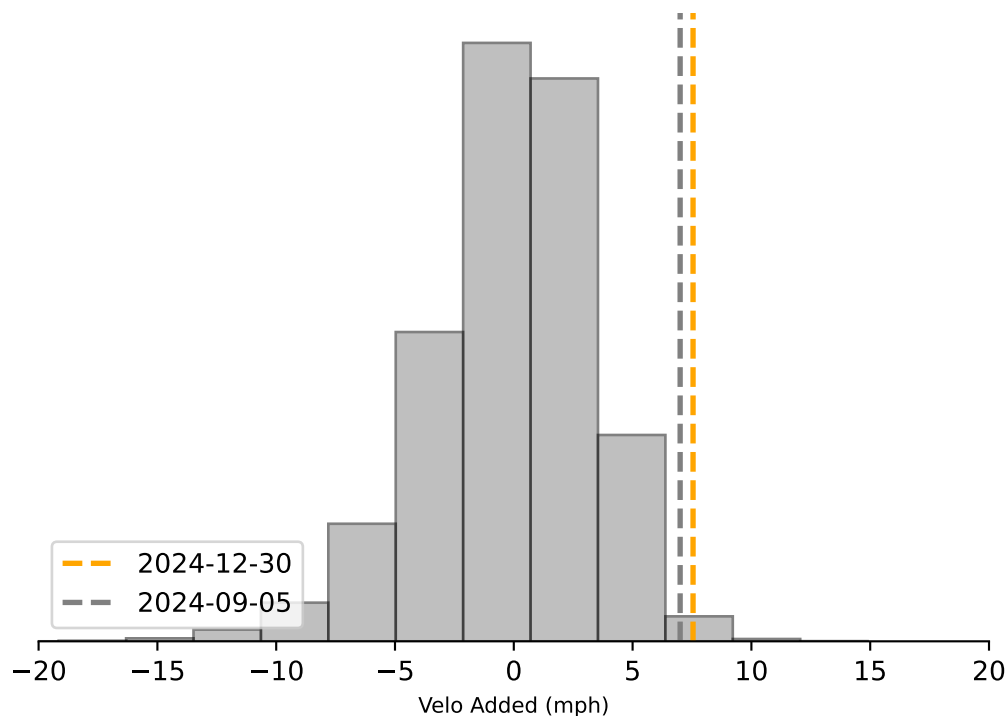
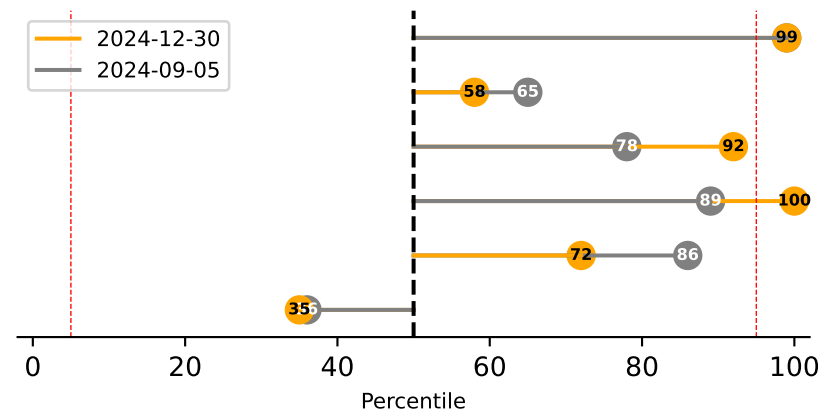


The arm action model provides a one-number estimate of how good the athlete's arm action is. Arm action is the second most impactful model for velocity & most impactful for velocity above expected.

The model summary is outlined below.

Per1mph is the estimated feature change needed, **on average**, for a 1mph change in velocity above expected.

Feature	Athlete	Median Elite	Importance	Per 1mph
Layback	212 deg (-0)	190 deg	high	5 deg
Elbow Extension Velo	2364 deg/s (-59)	2318 deg/s	med	203 deg/s
Shoulder Abduction at FP	98 deg (+7)	84 deg	med	10 deg
Scap Load at FP	89 deg (+26)	51 deg	med	16 deg
Shoulder Rotation Velo	4909 deg/s (-232)	4673 deg/s	med	588 deg/s
Elbow Flexion at FP	95 deg (-1)	102 deg	low	35 deg



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Bottom Left: Total velocity added across all model features.

Provides an estimate of how much velocity is added by the athlete's arm action, compared to the average arm action score